

Luis Navarro

Sr. AI & Backend Engineer | Python, LLMs, AI Agents | Fintech & Healthcare

SUMMARY

Senior **AI & Backend Engineer** with 10+ years of experience building **SaaS products, backend platforms, and production AI applications**. Strong background in **Python, FastAPI, LLMs, AI agents, RAG, APIs, PostgreSQL, and cloud infrastructure**, with hands-on work across **fintech, healthcare, and enterprise software**.

I've built and supported systems involving **document processing, vector search, structured data extraction, workflow automation, background jobs, third-party integrations, authentication, and human-in-the-loop AI workflows**. I focus heavily on the engineering around the model—making sure AI features are reliable, traceable, secure, and able to handle real production failures.

My experience also includes **React, Next.js, TypeScript, Node.js, C#, ASP.NET Core, AWS, Docker, and CI/CD**, which allows me to work across the full product when needed. I'm comfortable owning projects from **architecture and backend design through AI integration, deployment, and production support**.

WORK EXPERIENCE

Pixel Health (Acquired by Boston Children's Hospital)

Holyoke, MA

Senior AI & Backend Engineer

Feb 2022 – Nov 2025

- Designed and implemented **AI-powered healthcare solutions** that streamlined **clinical workflows**, automated **operational processes**, and enhanced **digital health** experiences.
- Built production-ready **LLM-powered applications, AI assistants, and Retrieval-Augmented Generation (RAG)** systems integrated with **enterprise healthcare platforms**.
- Developed scalable **Python backend services, RESTful APIs, and data pipelines** to support secure, reliable **AI applications** while ensuring **HIPAA-compliant** handling of sensitive healthcare data.
- Integrated **Large Language Models (LLMs), AI automation, and knowledge retrieval systems** into production environments to improve performance and user experience.
- Collaborated with **software engineers, clinicians, and product teams** to deliver production-grade **AI capabilities** that improved **automation, knowledge retrieval, and operational efficiency**.
- Focused on building **secure, scalable, and high-performance** backend systems using modern software engineering best practices.

Skills: Python, FastAPI, OpenAI, LangChain, LlamaIndex, RAG, pgvector, PostgreSQL, FHIR, AWS, Docker, GitHub Actions, CI/CD

American Renal Associates
Senior Full Stack Developer

Beverly, MA
Dec 2020 – Jul 2022

- Developed and maintained enterprise **healthcare applications** supporting clinical and business operations across a nationwide dialysis network. Designed and built scalable **backend services**, **RESTful APIs**, and system integrations using **C#** and **ASP.NET Core**, focusing on performance, security, and maintainability.
- Enhanced application features, optimized SQL Server databases, and improved data processing for business-critical healthcare systems. Collaborated with **product managers**, **QA engineers**, and cross-functional teams to deliver production-ready software through **Agile development**, **code reviews**, **testing**, and **CI/CD** practices. Resolved production issues, improved system reliability, and supported ongoing platform enhancements.

Skills: C#, ASP.NET, PostgreSQL, Redis, RESTful APIs, Docker, Jenkins, Git, CI/CD

IT Trailblazers
Backend Software Developer (Independent Contractor)

Boston, MA
Feb 2018 – Dec 2020

- Delivered backend solutions for enterprise client engagements using **Node.js** and **Express.js**. Designed and developed scalable **RESTful APIs**, **authentication services**, and business-critical backend components for web and SaaS applications. Built system integrations with **third-party** services and optimized database performance to improve application reliability and responsiveness.
- Worked with **cross-functional** teams to translate business requirements into maintainable technical solutions, troubleshoot production issues, and support feature enhancements throughout the software development lifecycle. Contributed to **Agile development**, **code reviews**, **automated testing**, **deployments**, and **ongoing production support** to ensure high-quality software delivery.

Skills: Node.js, Express.js, TypeScript, PostgreSQL, Redis, RESTful APIs, Docker, Jenkins, Git

EDUCATION

Associate's degree, Computer Science

North Shore Community College, Lynn, MA

KEY PROJECTS

- **AI Healthcare Assistant:** Developed an **LLM-powered healthcare assistant** using **Python, FastAPI**, and **Retrieval-Augmented Generation (RAG)** to automate clinical knowledge retrieval and streamline healthcare workflows, reducing information lookup time by **40%**.
- **Healthcare AI Platform:** Built a scalable **AI-powered healthcare platform** using **Python, FastAPI, PostgreSQL**, and **RESTful APIs**, integrating enterprise healthcare systems while ensuring **HIPAA-compliant** data handling and improving operational efficiency by **35%**.
- **SaaS Backend Platform:** Designed and developed a secure **SaaS backend** using **Python, Flask**, and **FastAPI**, implementing **JWT authentication, role-based access control (RBAC)**, and **RESTful APIs**, supporting thousands of concurrent users.
- **Enterprise API Integration Hub:** Developed a centralized integration platform using **Python, FastAPI**, and **Redis**, connecting **payment gateways, authentication providers, and third-party APIs**, reducing manual processing by **50%**.
- **Background Job Processing System:** Built an asynchronous task processing service using **Celery, Redis**, and **Python** for scheduled jobs, notifications, and report generation, improving backend processing performance by **45%**.
- **Enterprise Business Management System:** Developed backend services using **Node.js, Express.js, MongoDB**, and **REST APIs** for enterprise business applications, optimizing database performance and reducing API response times by **30%**.
- **AI Knowledge Retrieval System:** Implemented an **LLM and RAG-based** knowledge search platform using **Python, FastAPI, OpenAI APIs**, and **vector databases**, enabling intelligent document search and improving answer accuracy by **40%**.
- **React Dashboard & Analytics Portal:** Built a responsive **React** and **Next.js** dashboard integrated with **FastAPI** backend services, providing real-time analytics, user management, and reporting with secure authentication.

SKILLS

- **Languages:** Python, JavaScript, TypeScript, SQL, C#, ASP.NET
- **Backend Frameworks:** Flask, FastAPI, Node.js, Express.js
- **Frontend:** React.js, Next.js, HTML5, CSS3, Tailwind CSS
- **AI & Machine Learning:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, OpenAI API Integration, AI Automation

- **API Development:** RESTful APIs, API Integration, JWT Authentication, OAuth, Role-Based Access Control (RBAC), Webhooks
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis
- **Background Processing:** Celery, Redis, Asynchronous Task Processing, Scheduled Jobs
- **Cloud & DevOps:** AWS, Docker, Git, GitHub, CI/CD, Linux
- **Third-Party Integrations:** Payment Gateways, Authentication Providers, Cloud Services, External APIs
- **Software Development:** SaaS Development, Backend Architecture, System Design, Microservices, Database Design, Performance Optimization, Security Best Practices, Healthcare
- **Methodologies:** Agile, Scrum, Software Development Life Cycle (SDLC), Test-Driven Development (TDD), Code Reviews